

$$\hat{\rho} \left\{ \begin{array}{c} \text{---} \boxed{\hat{U}_A} \text{---} \\ \text{---} \boxed{\hat{U}_B} \text{---} \end{array} \right\} \hat{\rho}' \iff \hat{\rho}' = (\hat{U}_A \otimes \hat{U}_B) \hat{\rho} (\hat{U}_A \otimes \hat{U}_B)^\dagger$$